

BITS PILANI, DUBAI CAMPUS
DUBAI INTERNATIONAL ACADEMIC CITY, DUBAI.
THIRD YEAR - II SEMESTER 2019-20
DESIGN AND ANALYSIS OF ALGORITHMS (CS F364)
COMPREHENSIVE EXAMINATION (CLOSED BOOK)

Date: 04.06.2020

Max Marks: 80

Time: 8.30 to 11:30 am, 180 minutes

QUESTION 1 (8 MARKS)

Run the Maximum Program Stored Problem with $L=66$, $n=12$ and $\{n_1, n_2, n_3, n_4, n_5, n_6, n_7, n_8, n_9, n_{10}, n_{11}, n_{12}\} = \{26, 55, 8, 30, 28, 29, 2, 6, 5, 27, 7, 4\}$.

- A. Find the optimal number of programs stored with a disk of size 2L. (1 mark)
- B. Find the optimal number of programs stored using MPSP. (3 marks)
- C. Apply Absolute approximation algorithm with k value to be 1 and find one feasible solution. (4 marks)

QUESTION 2 (8 MARKS)

Given the 5*5 matrix of maze. Draw the state space tree using backtracking to find all the possible solution from source (S) to destination (D). The movement can be made in four directions only: left, right, top, down. No diagonal movement permitted. (4 marks)

Once after the completion of entire state space tree, find all the paths and the path that has minimum number of A. (4 marks)

S	×	A	×	B
A	B	A	×	B
×	B	B	B	A
×	×	×	B	D
A	A	A	×	×

QUESTION 3 (8 MARKS)

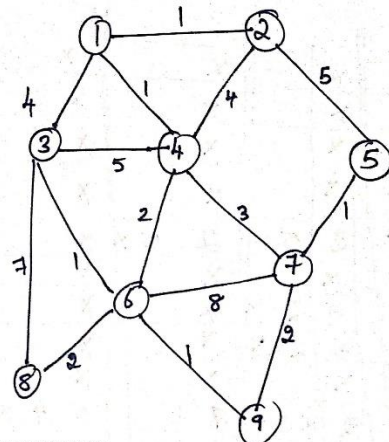
Mentioned below is the Prim's algorithm that you are aware of. Follow the algorithm and find the Minimum Spanning Tree of the graph. Draw the table with values

j	Near(j)
---	---------

 for every iteration.

```

1  Algorithm Prim( $E, cost, n, t$ )
2  //  $E$  is the set of edges in  $G$ .  $cost[1 : n, 1 : n]$  is the cost
3  // adjacency matrix of an  $n$  vertex graph such that  $cost[i, j]$  is
4  // either a positive real number or  $\infty$  if no edge  $(i, j)$  exists.
5  // A minimum spanning tree is computed and stored as a set of
6  // edges in the array  $t[1 : n - 1, 1 : 2]$ .  $(t[i, 1], t[i, 2])$  is an edge in
7  // the minimum-cost spanning tree. The final cost is returned.
8  {
9      Let  $(k, l)$  be an edge of minimum cost in  $E$ ;
10      $mincost := cost[k, l]$ ;
11      $t[1, 1] := k$ ;  $t[1, 2] := l$ ;
12     for  $i := 1$  to  $n$  do // Initialize near.
13         if  $(cost[i, l] < cost[i, k])$  then  $near[i] := l$ ;
14         else  $near[i] := k$ ;
15      $near[k] := near[l] := 0$ ;
16     for  $i := 2$  to  $n - 1$  do
17     { // Find  $n - 2$  additional edges for  $t$ .
18         Let  $j$  be an index such that  $near[j] \neq 0$  and
19          $cost[j, near[j]]$  is minimum;
20          $t[i, 1] := j$ ;  $t[i, 2] := near[j]$ ;
21          $mincost := mincost + cost[j, near[j]]$ ;
22          $near[j] := 0$ ;
23         for  $k := 1$  to  $n$  do // Update  $near[ ]$ .
24             if  $((near[k] \neq 0) \text{ and } (cost[k, near[k]] > cost[k, j]))$ 
25                 then  $near[k] := j$ ;
26     }
27     return  $mincost$ ;
28 }
```



(8 marks)

QUESTION 4 (7 MARKS)

Assume you run a multithreaded model of Fibonacci series with input as $n=5$.

A. Draw the complete Directed Acyclic Graph (DAG) representing the strands, nodes and their links. (3 mark)

B. Name the DAG with the different types of edges and nodes. (2 mark)

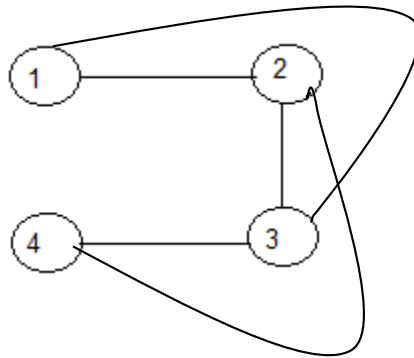
C. If the Fib (5) is executed with a single processor then it takes 22 time units to execute. Calculate the parallelism in this case. (2 mark)

D. Given below is the algorithm of parallel and multithreaded Fibonacci for your reference.

```
P-FIB( $n$ )
1  if  $n \leq 1$ 
2    return  $n$ 
3  else  $x = \text{spawn P-FIB}(n - 1)$ 
4        $y = \text{P-FIB}(n - 2)$ 
5       sync
6       return  $x + y$ 
```

QUESTION 5 (8 MARKS)

Given the graph



Given the graph $G=(V,E)$. Find the max clique of G . Construct another graph G' that contains 3 copies of graph G , with the same edges.

Hence $G'=(V',E')$ where $V'=12$ nodes i.e. $\{1,2,3,4\},\{1',2',3',4'\},\{1'',2'',3'',4''\}$.

In G' , connect every individual vertex of each copy of the graph G is to all the vertices of the other two copies of the graph G . Say 1 is connected to $1',2',3',4',1'',2'',3'',4''$ and similarly the other vertices.

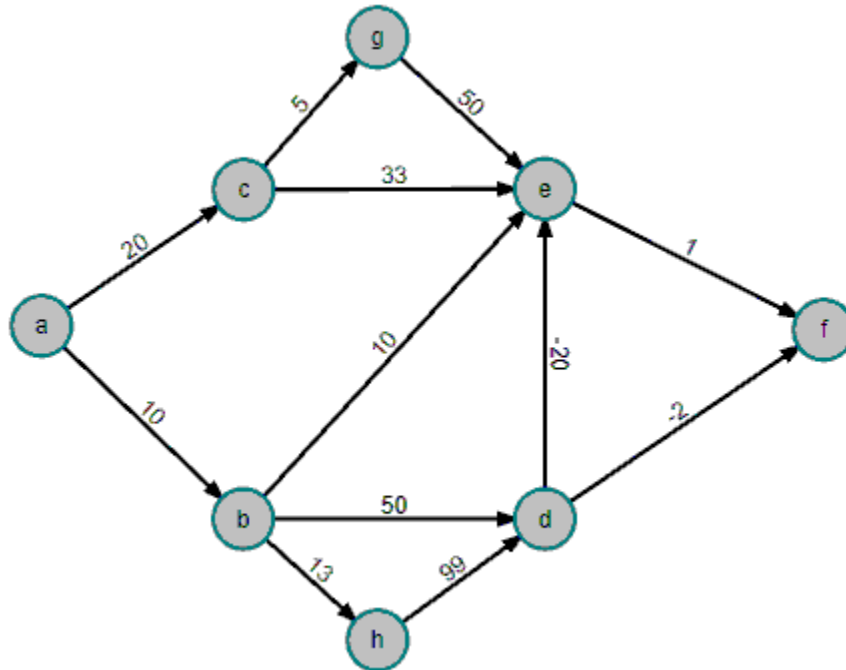
A. Draw the Graph G' using the above constraints. (4 marks)

B. Relate the max clique of graph G and G' . (3 marks)

C. Apply absolute approximation algorithm with k value as 5 on the graph G' and find the feasible clique. (1 mark)

Section -2(30 Marks)

1. Consider the following given graph

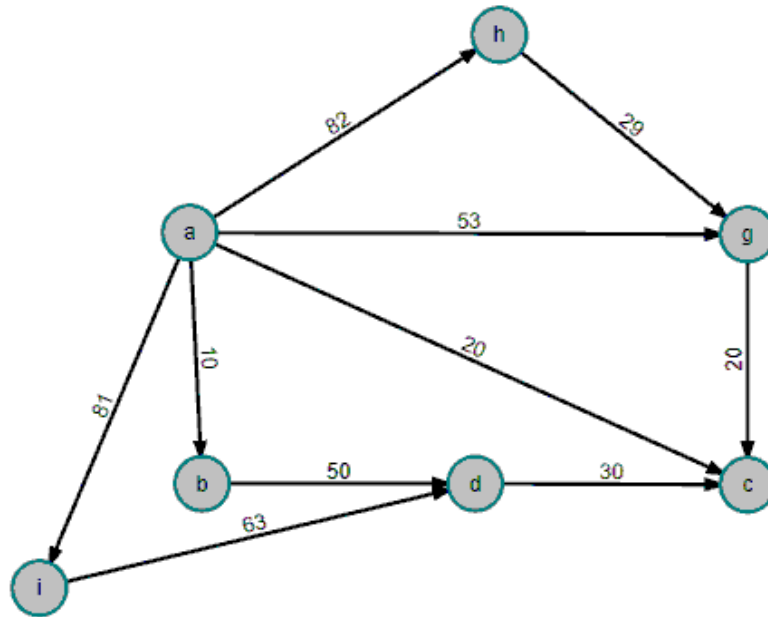


Assume the start node is 'a'. Compute the distance matrix representing the single source all pair distance matrix using the BellmanFord algorithm. Show all the calculations of the distance between the source to the nodes using a recurrence relation. Represent the computed distance matrix in the following form only.

k	a	b	c	d	e	f	g	h
1								
2								
3								
4								
5								
6								
7								

[7]

2. Write the algorithm for augmenting a path in the residual network. For the following given network source is 'a' and sink is 'c', consider a path with the flow of 20 from node **a to i to d to c**, write the residual graph.



[8]

3. Write a non-deterministic algorithm to perform sorting of all the rows of a matrix into ascending order

[7]

4. Consider the following objective function and constraints,

Minimize $11x_1 + 3x_2 - 10x_3$

Subjected to

$$13x_2 + 8x_3 = 6$$

$$-5x_1 + 9x_2 + 2x_3 \geq 20$$

$$x_3 \geq 0$$

- i. Convert the above linear programming system into standard form
- ii. Using the standard form of the linear programming system convert to slack form.

[8]

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THIRD YEAR - II SEMESTER 2019-20
DESIGN AND ANALYSIS OF ALGORITHMS (CS F364)
TEST -2 (OPEN BOOK) – QUESTION PAPER AND ANSWER KEY
ONLINE EXAM IN TEST FOR GEEKS

DATE: 09.04.2020
TIME: 8.30 TO 9.15 AM, 50 MINUTES

MAX MARKS: 40
WEIGHTAGE: 40 %

There are total 11 questions and each Question carries equally 3.63 i.e. (40/11) marks

QUESTION 1

We need to write a non-deterministic algorithm to generate a spanning tree T from the graph $G=(V,E)$. Here $V=\{v/v \text{ is a vertex belongs to } V\}$, $E=\{(u,v)/u \text{ and } v \text{ are vertices in } V\}$, $W=w(u,v)$ is a weight matrix such that if there is an edge (u,v) there $w(u,v)$ is a real value otherwise $w(u,v)$ is infinity. The tree is represented as T list of the edge as $T=\{(u,v)/ (u,v) \text{ belongs to } E\}$.

Identify which of the following will be the most appropriate use of function choice, and conditions to that will be satisfied when T is a minimum spanning tree

1. ☐ Choice (True/False) [Choice selects True or False]
2. ☐ Choice() [Choice selects a random number between $-\infty$ to ∞]
3. ☒ Choice(v) [Choice selects one vertex from the set V]
4. ☐ Choice(e) [Choice selects one edge from the edge set E]
5. ☐ T has an edge (u, v) such that (u,v) not belongs to E
6. ☒ T has an edge (u, v) such that (u,v) belongs to E
7. ☒ T has an edge (u, v) such that (u,v) is minimum among all the incident edges from node u or minimum among all the incident edges from node v
8. ☐ T has an edge (u, v) such that (u,v) creates a cycle in the Tree T

QUESTION 2

Which of the followings are true about P and NP

1. ☒ P: the class of decision problems that have polynomial-time deterministic algorithms.
2. ☒ NP: the class of decision problems that are solvable in polynomial time on a nondeterministic machine
3. ☐ P: the class of decision problems that have polynomial-time nondeterministic algorithms
4. ☒ P: the class of decision problems with time complexity for examples $O(3x^2+2y)$, $O(2x+y+z)$
5. ☐ NP: the class of decision problems that are solvable in polynomial time on a deterministic algorithm
6. ☐ NP: the class of decision problems that are solvable with Non Polynomial-time algorithm using a deterministic algorithm

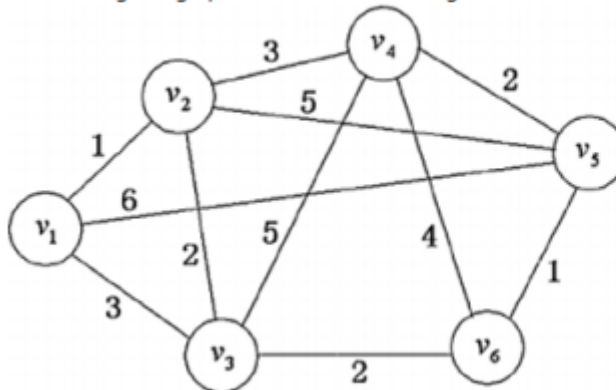
QUESTION 3

Consider satisfiability with at most three literals per clause reduces to chromatic number decision problem and vice versa. In this reduction, problem find the which of the following statements are true

1. ☒ Here we have given F a CNF with at most three literals per clause. There are r number of clauses in F. There are x_i between 1 to n variables.
2. ☒ An examples for CNF with at most three literals per clause is $(x_1 \wedge x_3) \vee (\neg x_1) \vee (\neg x_2 \wedge \neg x_3)$
3. ☒ The graph is defined and constructed from F. $G=(V,E)$, V is vertex set and E is edge set. For a CNF $F=(x_1 \wedge x_2 \wedge x_3) \vee (\neg x_1 \wedge x_2 \wedge \neg x_3) \vee (x_1 \wedge \neg x_2 \wedge x_3)$, total number of vertices in G is 12
4. ☐ The graph is defined and constructed from F. $G=(V,E)$, V is vertex set and E is edge set. For a CNF $F=(x_1 \wedge x_2) \vee (\neg x_1 \wedge x_2)$. The edge list has following edges $\{(x_1, \neg x_2), (\neg x_1, x_2), (x_1, x_2)(x_1, C1), (C2, C1), (y1, y2), (x1, y1), (x2, y2)\}$
5. ☐ Consider a CNF F with r number of clauses and consists of n variables. The graph G generated for F with $G=(V,E)$. Here E is the edge set and the edge set E has exactly $n*n*n*r$ number of edges.

QUESTION 4

For below-given graph which of the followings make the node cover



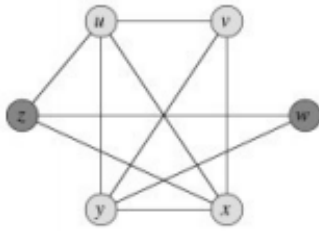
1. ☐ {v1,v4,v6}
2. ☒ {v1, v3,v6,v4,v2}
3. ☐ {v2,v5,v6,v4}
4. ☒ {v2,v5,v6,v3}

QUESTION 5

Consider 5 jobs (j_1, j_2, j_3, j_4, j_5) with the processing time as $(t_1, t_2, t_3, t_4, t_5) = (5, 7, 12, 6, 3)$ and there are two processors P1 and P2. The non-preemptive scheduling S of these jobs on the processors is done as P1 is assigned jobs in order (j_1, j_2, j_4) and P2 is assigned (j_3, j_5). When P1 does finish processing of all its jobs? When P2 does finish processing of all its jobs? What is finish time for schedule S?

1. ☒ P1 finishes processing of all its jobs at 18
2. ☐ P1 finishes processing of all its jobs 15
3. ☒ P2 finishes processing of all its jobs 15
4. ☐ P2 finishes processing of all its jobs 18
5. ☐ P1 finishes processing of all its jobs 25
6. ☐ P2 finishes processing of all its jobs 28
7. ☒ Finish time for schedule S is 18
8. ☐ Finish time for schedule S is 15
9. ☐ Finish time for schedule S is 28

QUESTION 6



Q1. Find the max-clique for the following graph?

Q2. Find the clique by assuming that you apply absolute approximation with value of k to be 1.

Choose the possible solutions from below that satisfies the above question Q1 and Q2

1. ☒ $\{u,v,x,y\}$ and $\{u,v,y\}$
2. ☒ $\{u,v,x,y\}$ and $\{u,x,y\}$
3. ☒ $\{u,v,x,y\}$ and $\{u,x,z\}$
4. ☐ $\{u,v,y\}$ and $\{u,v\}$
5. ☐ $\{u,v,y\}$ and $\{u,v,y\}$
6. ☒ $\{u,v,x,y\}$ and $\{u,v,x,y\}$

QUESTION 7

Q1. Let graph $G=(V,E)$ and we construct graph $G'=(V',E')$ such that G' is made up of 3 copies of graph G . These $k+1$ copies of G is connected to form a complete graph such that there is every edge between all the vertices (no self loops)
(Choose from the first five options)

Q2. Which absolute approximation algorithm do you prefer to be for a planar graph assuming that it should give the best feasible solution?

(Choose from the last five options)

1. ☐ If the maximum clique size in graph G is ' m ' then ' m ' will be the maximum clique in G' .
2. ☒ If the maximum clique size in graph G is ' m ' then ' m ' will be the minimum clique in G' .
3. ☒ If the maximum clique size in graph G is ' m ' then ' $3m$ ' will be the maximum clique in G' .
4. ☐ If the maximum clique size in graph G' is ' n ' then ' n ' will be the maximum clique in G .
5. ☒ If the maximum clique size in graph G' is ' n ' then a specific subset of n will form a maximum clique in G .
6. ☐ 2-approximation
7. ☒ 1-approximation
8. ☐ 4-approximation
9. ☐ 3-approximation
10. ☐ The approximation value will depend on the number of nodes and the connection pattern between the nodes.

QUESTION 8

Assume a hardware engineer and want to fix small disks into big disk of size 15 kb. Name of the small disks and their size is given below:

Constraint: The smaller disk has to be allocated as a whole to bigger disk and cannot be broken into pieces.

Small Disk	A	B	C	D	E	F	G	H	I	J	K
Size	10	5	14	13	8	7	6	2	4	12	4

Find the number of bins needed to accommodate the smaller bins into bigger bins using

1. Find the number of bins needed to accommodate the smaller bins into bigger bins using First Fit
2. Find the number of bins needed to accommodate the smaller bins into bigger bins using FFD
3. Write the name of the small disks filled in 4th bin using First Fit
4. Write the name of the small disks filled in 2nd bin using FFD
5. Write the name of the small disks filled in 5th bin using First Fit
6. Identify in which bin the item 'd' is put using FFD.

Choose the right combination of answers from below options

1. ☐ Q1 (7) Q2 (6) Q3 (h,d) Q4 (e,f) Q5 (g,k) Q6 (3)
2. ☐ Q1 (6) Q2 (6) Q3 (e,f) Q4 (h,d) Q5 (g,i) Q6 (2)
3. ☒ Q1 (6) Q2 (6) Q3 (e,f) Q4 (h,d) Q5 (g,i,k) Q6 (2)
4. ☐ Q1 (7) Q2 (7) Q3 (e,f) Q4 (h,d) Q5 (g,i,k) Q6 (2)

QUESTION 9

Given the graph G with 5 vertices $\{1,2,3,4,5\}$ and with following edges $E=\{1-2,1-3,1-4,2-3,3-4,2-5,4-5\}$

Construct the graph $G'=\{V,E'\}$ from graph $G=\{V,E\}$ with few changes as given below.

Connect all the vertices in graph G' forming a complete graph (no self-loop) such that number of edges in G' will obviously be more than G .

Allocate weightage to the edges in G' according to the following conditions

1. Assign the weight as 1 for those edges that were in graph G
2. Assign the weight as 'k' for all the other edges that were not in G . Run the TSP algorithm in graph G' . Study the graph G' .

Q1. Run the travelling salesman problem with graph G' as input. What should be the optimal value of shortest path that you should get in G' to confirm that G has Hamiltonian cycle.

(Choose the answer from first five options)

Q2. Assume the value of k is 4 in G' and the edge 2-3 is removed in G' . You run the Shortest path algorithm and obtain the value as $1+k$. Do you think G has Hamiltonian cycle.

(Choose the answer from last two options)

1. ☐ 7

3. ☐ 10

2. ☐ 8

4. ☐ 6

5. ☒ 5

6. ☒ Yes

7. ☐ No

QUESTION 10

Q1.

1. Bin packing can be reduced to 0/1 Integer programming
2. Bin packing can be reduced to fractional Integer programming
3. Bin packing is not an NP hard problem

(Among the first four options, which is correct?)

Q2. Given the 0/1 knapsack problem, $m = 100$

| SNo | Profit | Weight

| 1 | 110 | 50

| 2 | 90 | 60

| 3 | 100 | 40

| 4 | 120 | 30

Apply 0.2-approximate algorithm with k value to be 0.2 and find the number of feasible solutions available?

(Among the last four options, which is correct?)

2. ☐ 1- false 2- false 3- false

8. ☐ 3

1. ☐ 1- true 2- true 3- false

3. ☐ 1- true 2- true 3- true

4. ☒ 1- true 2- false 3- false

5. ☐ 4

6. ☒ 5

7. ☐ 6

QUESTION 11

Q1. Which of the following statement is true with approximation algorithms?

(Choose the right answer from the first four options)

Q2. What is the maximum number of colors needed to color a planar graph.

(Choose the right answer from the last four options)

1. ☒ AL takes sub exponential time or polynomial time
2. ☐ AL can be designed for all the NP hard problems
3. ☒ Heuristics can ease in reducing the NP hard algorithms to AL
4. ☒ Probabilistic good algorithms always generate optimal solution
5. ☐ 3
6. ☐ 2
7. ☒ 4
8. ☐ 1

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THIRD YEAR - II SEMESTER 2019-20
DESIGN AND ANALYSIS OF ALGORITHMS (CS F364)
TEST -1 (CLOSED BOOK)

DATE: 27.02.2020
TIME: 8.30 TO 9.20 AM, 50 MINUTES

MAX MARKS: 40
WEIGHTAGE: 40%

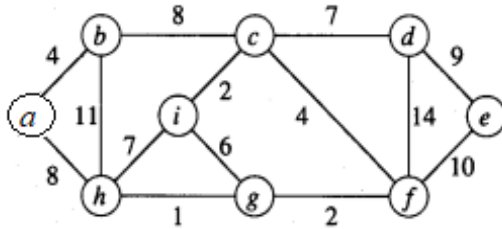
1. Write a divide and conquer algorithm to find the smallest and second smallest elements in an array. For example $arr[] = \{21, 130, 2, 10, 4, 3\}$. Smallest is 2 and second smallest is 3.
[6]
2. We have given a knapsack (bag) with capacity 152 and there are 10 items ($i_1, i_2, i_3 \dots i_{10}$). The weights in kg of 8 items are: (8,19,21,33,11,14,2,45). The profits for each item are: (111,82,91,100, 24,37,18,9). Using greedy method write the solution for the followings
 - a. Knapsack is filled in order of highest to lowest profit. Write the profit obtained and items which are selected
 - b. The ratio of profit by weight is computed. Knapsack is filled in order of highest to lowest this ratio. Write the profit obtained and items which are selected
[8]
3. We have given an array {21, 17, 9, 8,11, 14, 7}. Perform the merge sort algorithm on this array and write steps showing working of Merge sort algorithm. Also show the values of variables used in the merge sort algorithm.
[6]
4. Apply job scheduling to the jobs (j_1 to j_7) with profits={3,5,20,18,1,6,30} with deadline={4,1,4,1,3,2,2}. Do an optimal assignment to complete executing the jobs without collision or overlapping.
[6]
5. Let $G=\{V,E\}$ be a weighted directed graph. Each pair of vertices $\{v_i, v_j\}$ is connected with weight w_{ij} .
 - a. Write an algorithm to find the maximal distance from every node to every other node in the graph. Mention the formulae to be used. The graph is permitted to have to have a cycle and negative weights. Be concerned that you don't get caught in loop, infinitely. Include this constraint in the algorithm too.
[6]

- b. Mention the Worst case complexity of the algorithm. Suggest an approach to optimize this algorithm and reduce the time complexity further.

[2]

6. Using Prim's greedy algorithm obtain minimum spanning tree from the given graph. Draw the step by step representation of reducing the graph.

[5]



Mention complexity of the algorithm.

[1]

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THIRD YEAR - II SEMESTER 2019-20
DESIGN AND ANALYSIS OF ALGORITHMS (CS F364)
QUIZ – QUESTION PAPER AND ANSWER KEY
ONLINE EXAM

DATE: 19.03.2020

TIME: 01.00 PM TO 01:20 PM, 20 MINUTES

MAX MARKS: 20

WEIGHTAGE: 20 %

There are 10 questions and each question takes 2 marks

1. What is the correct definition of NP time
 - a. NP stands for non-polynomial algorithm time
 - b. NP stands for non-deterministic polynomial algorithm time**
 - c. NP stands for non-computational polynomial algorithm time
 - d. NP stands for non-performing polynomial algorithm time
 - e. Both a and b
 - f. both c and d
2. Consider a graph G consists of n nodes. Only three subgraphs G1, G2 and G3 possible to generated and they have the size k1, k2, and k3, such that $k_1 > k_2$ and $k_2 > k_3$. Property of G1, G2 and G3 is such that all the nodes are connected with each other. What is size of maximum clique for the graph G in this case?
 - a. k1**
 - b. k2
 - c. k3
 - d. none
3. A non-deterministic algorithm uses 3 functions. Mention the name of these three functions
 - a. Random(x)
 - b. Choice ()
 - c. Success()
 - d. Failure()
 - e. Function a, b, c
 - f. Function b, c, d**
 - g. Function a,b,d
4. Which of the following Boolean expression is in CNF?
 - a. $(x_1 \text{ or } x_2 \text{ or } x_3) \text{ and } (x'_1 \text{ or } x'_2) \text{ and } (x_1 \text{ or } x'_2 \text{ or } x'_3)$**
 - b. $(x_1 \text{ and } x_2 \text{ and } x_3) \text{ or } (x_1 \text{ and } x'_2 \text{ and } x_3) \text{ or } (x'_1 \text{ and } x_2)$
 - c. $(x_2 \text{ and } x_3) \text{ and } (x_1 \text{ and } x'_2 \text{ or } x_3) \text{ or } (x_1 \text{ and } x_3)$
 - d. $((x_2 \text{ or } x_3) \text{ or } (x'_1 \text{ and } x'_2 \text{ and } x'_3) \text{ and } x_2) \text{ or } (x_1 \text{ and } x'_3)$
5. Consider a graph which has $V = \{A, B, C, D, E, F\}$ and the edge set $E = \{(A, B), (A, C), (A, D), (C, B), (B, E), (D, E), (D, F)\}$. Find the minimum number of colors required to color this graph
 - a. 2
 - b. 3**
 - c. 4
 - d. 5

6. Let $w=\{5,7,10,15,18,20\}$ and $m=35$. Find all the subsets of w that sums up to m . Choose the number of subsets that you obtain.

- a. 1
- b. 2
- c. 3**
- d. 4

7. What is the condition that checks the diagonal clash in N-Queens problem? Let (a,b) and (c,d) refers to the index of placing the queens.

$$a-b=c-d$$
$$a+b=c+d$$

- a. Only A
- b. Only B
- c. Both A and B**

8. Let the profit of unbreakable items be $\{1,1000\}$ and the weight of the items be $\{1000,1\}$ in kg. The maximum capacity of the knapsack bag be 1000 kg. Choose one optimal solution and one feasible solution to this problem. Calculate the absolute difference of their profit.

- a. 1000
- b. 1
- c. 999**
- d. 900

9. Map the algorithms with their complexities

- A. Greedy Knapsack
- B. Fractional Knapsack
- C. Brute force N-Queens algorithm
- D. Minimum Spanning Tree

- 1. $O(N^N)$
- 2. $O(N^2)$
- 3. $O((N+E) \log N)$
- 4. $O(N \log N)$

- a. A-1, B-2, C-3, D-4
- b. A-4, B-2, C-1, D-3**
- c. A-4, B-3, C-2, D-1
- d. A-3, B-2, C-1, D-4

10. How many subsets can be generated from 12 elements?

- a. 12
- b. 144
- c. 4096**
- d. 1728